

Design Rationale: Mini Library Management System

Data Structure Choices

Reasons for the following Data Types

1. Dictionary for Books (``books = {}``)

- Fast ISBN-based Lookup: Dictionaries provide $O(1)$ average time complexity for lookups, which is crucial for frequent operations like borrowing, returning, and updating books by their unique ISBN-

O (1) Time Complexity: This is just a fancy way of saying "**One Step**". No matter how many books are in the dictionary, finding one by its ISBN takes essentially one single step. The time it takes does **not** depend on the size of the collection.

Crucial for Frequent Operations: In a library system, the most common actions are:

"Can I borrow *this specific book* (by ISBN)?"

"I want to return *this specific book* (by ISBN)."

"Update the details for *this specific book* (by ISBN)."

Using a dictionary for these tasks is like having that smart librarian. It's incredibly fast and efficient, which is vital for a system that needs to be responsive.

- Natural Key-Value Mapping: The ISBN serves as a perfect natural key that maps to book details (title, author, genre, copies)
 - Prevents Duplicates: Dictionary keys are inherently unique, preventing accidental duplicate ISBN entries
 - Efficient Updates: Easy to update specific book details without searching through entire collections

2. List for Members (``members = []``)

- Simple Sequential Access: Member operations (adding, searching) are less frequent than book operations
- Flexible Structure: Lists allow for easy iteration when displaying all members or searching by various criteria
- Natural Ordering: Maintains insertion order which can be useful for administrative purposes
- Memory Efficiency: For the expected scale of a university library, linear search performance is acceptable

3. Tuple for Genres (``VALID_GENRES = (...)``)

- Immutability: Prevents accidental modification of valid genres during runtime
- Data Integrity: Ensures genre validation remains consistent throughout the application
- Memory Efficient: Tuples are more memory-efficient than lists for fixed data
- Performance**: Faster iteration for validation checks